Experiment 03

Title

House Price Prediction using Linear Regression

SAP ID:	590013857
Dataset:	https://www.kaggle.com/datasets/mohamedafsal007/house-price-dataset-of-india/data

AIM

To build a machine learning model using Linear Regression to predict house prices based on various features and evaluate its performance.

THEORY

Linear Regression is a supervised learning algorithm used to model the relationship between dependent and independent variables.

Key concepts:

- **Regression**: Predicting continuous values
- **Feature Scaling**: Standardizing features improves performance
- **Model Evaluation**: Using MAE, MSE, RMSE, and R^2
- **Residual Analysis**: Helps validate model assumptions
- **Feature Importance**: Coefficients indicate influence of features

Commands/Code Used

```
# Key steps of the implementation

# Load dataset
df = pd.read_csv("House Price India.csv")

# Preprocessing
df = df.drop_duplicates()

# Feature selection
X = df.drop("Price", axis=1)
y = df["Price"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y)

# Scaling
scaler = StandardScaler()
```

```python
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Model
model = LinearRegression()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)
```

Results

1. Actual vs Predicted Prices

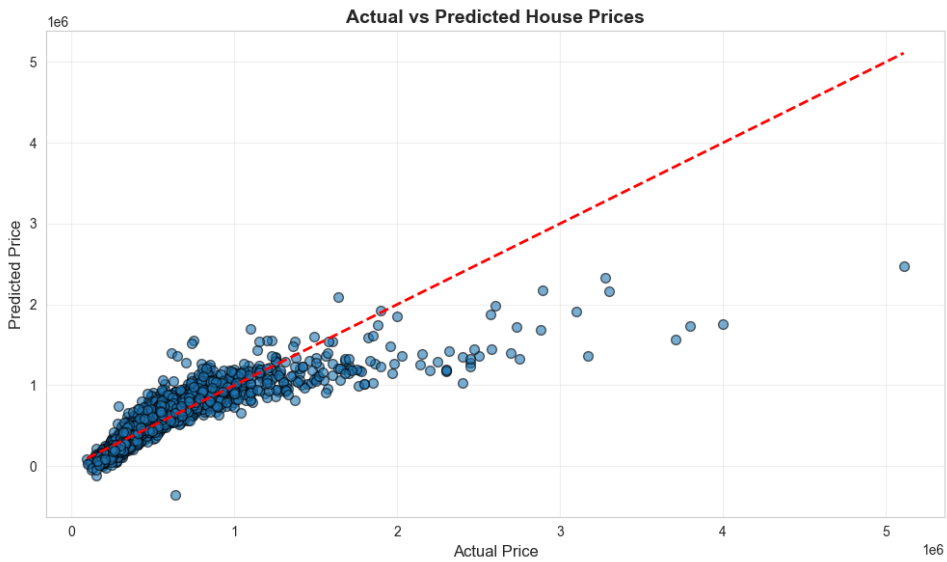

Figure 1: Actual vs Predicted house prices

2. Residual Analysis

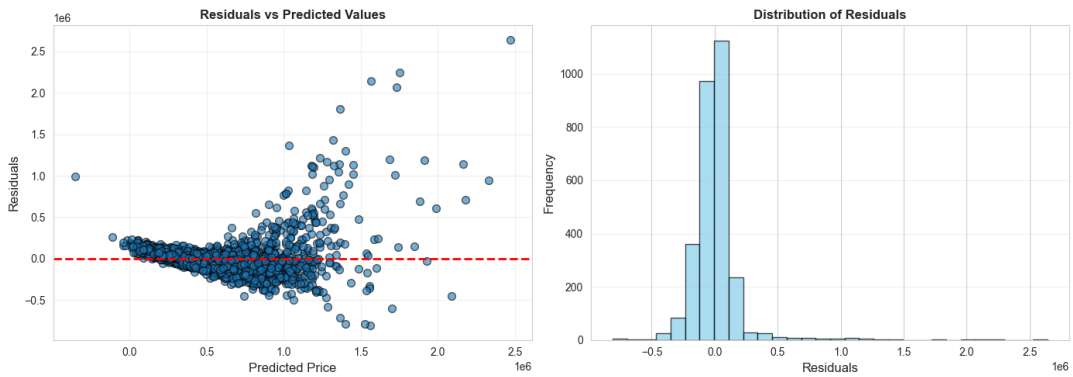

Figure 2: Residuals vs predicted values and residual distribution

3. Feature Importance

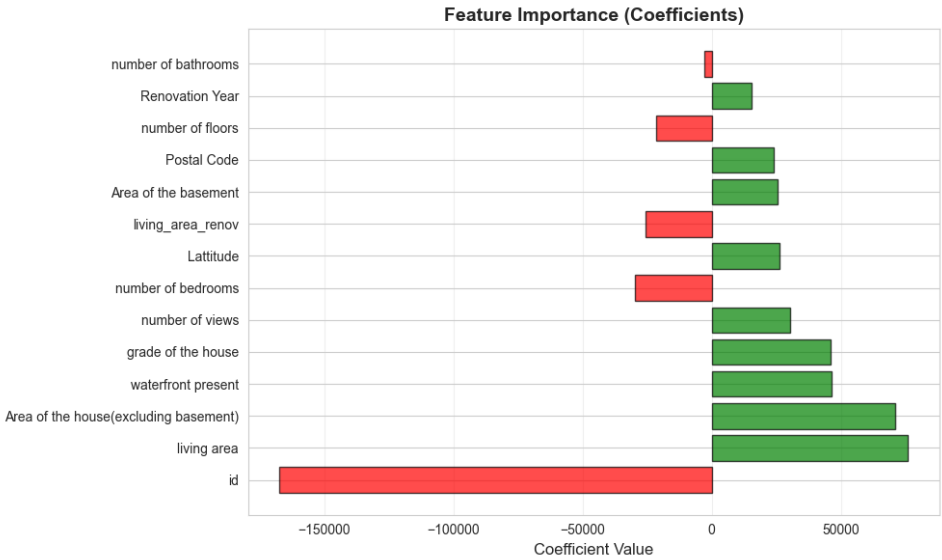

Figure 3: Feature importance based on regression coefficients

Conclusion

In this experiment, we successfully implemented a Linear Regression model to predict house prices. The model demonstrated a reasonable ability to capture relationships between features and target values.

Residual analysis showed that most errors are centered around zero, indicating a fairly good fit. Feature importance analysis revealed which attributes significantly influence house prices.

Overall, this experiment highlights the importance of preprocessing, feature selection, and evaluation in building reliable machine learning models.